

DU TUAN AN (俞俊安)

MANUFACTURING & INDUSTRIALIZATION ENGINEER

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github.com/dutuanan96 | linkedin.com/in/an-du-010b83420 | Portfolio: github.com/dutuanan96/engineering-portfolio

~150 SKUs owned	20+ OEM suppliers	22 / 14d SKUs delivered	15 / 7d SKUs rebuilt	2 / 2 IKEA GO/NOGO	4+ yrs Engineering
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PROFESSIONAL SUMMARY

Manufacturing and structural engineer with 4+ years of hands-on experience across NPI, OEM/ODM product development, supplier industrialization, CAD/CAM, tooling/fixture design, quality, BOM/PDM and engineering automation. Promoted from factory operations into engineering and process leadership, then moved into brand-side supplier engineering and manufacturing automation. *Uses AI coding agents to translate engineering requirements into practical tools, while owning workflow definition, testing, validation and deployment rather than traditional software coding.*

CORE COMPETENCIES

Manufacturing Engineering • NPI • Product Industrialization • Supplier Engineering • Product Development • Siemens NX • AutoCAD • CAD/CAM • Fixture & Jig Design • CNC • BOM • PDM • Packaging Engineering • Quality Control • AI-Assisted Engineering Automation

PROFESSIONAL EXPERIENCE

Manufacturing Automation Engineer | Jintai Furniture (金汰家具)

May 2026 - Present | Binh Duong, Vietnam

OEM/ODM Product Development | SONGMICS | Snow Shovel Product Line

- Delivered complete 3D models, production-ready 2D drawings, BOMs, assembly drawings and manufacturing documentation for 22 SKUs within 14 days.
- Generated complete CNC cutting drawings and production files for entire product lines within 24 hours; completed new-product engineering, prototype optimization and customer sample approval within 3 days.
- Defined the architecture, data rules and engineering workflows for an internal AI-integrated PDM system, then used Codex,
- Applied AI-assisted CAD/PDM automation to repetitive engineering-data, document and validation workflows, reducing manual effort and improving consistency across selected tasks.

Senior Structural Engineer | Beijing Huaqing Technology (北京华青科技)

Mar 2025 - Apr 2026 | Binh Duong, Vietnam

Brand-side engineering for Rolanstar, Greenstell and Seventable products manufactured in Vietnam

- Owned structural engineering and lifecycle maintenance for nearly 150 SKUs, including V2/V3 revisions, 3D model updates, production drawings and manufacturing improvements.
- Evaluated 20+ OEM suppliers in Vietnam and directly owned engineering deployment for 10+ core production suppliers.
- Reconstructed missing 3D product data through on-site measurement and CAD modeling; completed engineering data for 15 SKUs within one week to meet critical prototype deadlines.
- Held engineering authority for prototype sign-off and drop-test pass/fail decisions before Purchasing released production orders.
- Developed 3D packaging concepts and SOPs, maintained engineering BOMs, created assembly instructions/carton markings, and coordinated suppliers through prototype build, validation and production readiness.

Process Engineering Supervisor | 工艺主管 | Henglin Home Furnishings (恒林家居股份有限公司)

Jul 2024 - Mar 2025

IKEA office chair, sofa and bed programs

- Led two annual IKEA GO/NOGO factory assessments as the primary technical coordinator, worked directly with IKEA auditors, drove corrective actions and successfully passed both assessments.
- Translated IKEA PDOC requirements into factory processes, SOPs, inspection criteria and production controls covering drawings, testing, packaging and manufacturing.
- Designed jigs, fixtures and semi-automated equipment, converting selected 2-3 person operations into single-operator workflows; managed tooling development from CAD concept through validation and deployment.
- Led NPI from CAD design and CNC programming through prototype preparation, packaging and production implementation; managed a 2-person process engineering team and concurrently oversaw a 10-person metal-workshop QC team.

PROFESSIONAL EXPERIENCE - CONTINUED

Engineering Assistant | 工程师助理 | Henglin Home Furnishings (恒林家居股份有限公司)

Aug 2023 - Jun 2024

- Created 3D models and 2D production drawings for furniture and metal components; programmed Yaskawa welding robots and prepared Longxin CNC laser-cutting programs for production.
- Supported BOM creation, material requirements, packaging, prototype development and manufacturing documentation.

Production Data & Operations Support | Henglin Home Furnishings (恒林家居股份有限公司)

Mar 2022 - Jul 2023

- Maintained production data, calculated worker wages, supported daily factory coordination and translation, and applied 2D/3D CAD knowledge to assist the engineering team.
- Progressively took on technical assignments and equipment training, leading to a formal transfer into the engineering team in August 2023.

SELECTED ENGINEERING AUTOMATION & AI PROJECTS

AI-Integrated PDM & Engineering Change System *Professional Engineering Project*

github.com/dutuanan96/bom-viewer-sync

- Defined manufacturing requirements, PDM domain rules and workflows for products, materials, BOMs, revisions, engineering documents and controlled data changes in a browser-based Viewer/Admin system.
- Used AI coding agents to implement and iterate the system; personally tested workflows, identified failures, corrected engineering logic and validated results against real manufacturing scenarios.
- Designed an evidence-first AI flow: canonical PDM retrieval, structured change proposals, before/after diff, graph/domain validation, categorized risk warnings, engineer review and explicit final save rather than unrestricted AI mutation.
- Covered product/material search, where-used, revision comparison, BOM analysis, engineering-drawing analysis and controlled Admin proposals while keeping the Viewer read-only.
- Validated multi-turn AI behavior with deterministic regression checks and a multilingual workflow corpus containing 60 semantic transitions and 300 Vietnamese, Chinese and mixed-language variants.
- Demonstrated controlled engineering-change analysis for a 60 mm -> 100 mm paper-card specification update, including affected/shared BOM relationships, revision risk and approval before release.

Siemens NX12 Engineering Export Automation Dashboard *Professional CAD Automation Project*

github.com/dutuanan96/nx12-export-dashboard

- Defined and validated an AI-assisted Siemens NX batch-export workflow covering PRT -> PDF, PRT -> STEP (AP214), PRT -> DWG and STP/STEP -> IGES conversion through NX Journal/run_journal.
- Automated repetitive NX steps including Drafting-mode switching, multi-sheet drawing export, automatic output-folder creation and post-export file verification.
- Added practical production safeguards including log cleanup/retry, timeout handling, IGES output-size checks and DWG warnings for PRT files above the configured size threshold.

Furniture Manufacturing AI Agent - Google x Kaggle 5-Day AI Agents Capstone *Capstone Project, 2026*

github.com/dutuanan96/furniture-agent-capstone

- Designed a multi-agent BOM assistant using Google ADK with a root coordinator and specialized BOM and product-comparison agents, plus tool-based memory and Excel export.
- Structured tool use through AgentTool/FunctionTool patterns and an MCP-based tool layer, separating deterministic BOM parsing/comparison from model responses.
- Built the project around 22 furniture products and 44 bilingual BOM files, with Vietnamese/Chinese interaction, same-language response rules and read-only safeguards to prevent accidental data mutation.

ROLE ACROSS AI PROJECTS: engineering requirements, domain/workflow design, AI-agent orchestration, testing, validation and deployment. Implementation was produced with AI coding agents; this CV does not claim traditional software-development expertise.

TECHNICAL & DIGITAL SKILLS

Manufacturing

NPI, OEM/ODM, supplier engineering, product industrialization, process engineering, fixture/jig design, CNC, packaging, QC

AI / Digital

Codex, Antigravity, Hermes Agent, AI-assisted engineering automation, PDM workflow design, OCR-assisted tools, document/data validation

Languages

Vietnamese - Native; Chinese - Full Professional; English - Basic working proficiency / technical documentation

Education

TDMU - Thu Dau Mot University; Engineering / Industrial Management Studies

CAD / Engineering

Siemens NX, AutoCAD, 3D modeling, 2D production drawings, CAD/CAM, BOM, PDM, engineering documentation

Shopfloor

Yaskawa robotic welding, Longxin CNC laser cutting, metal fabrication, tooling, prototype build and validation

Certification

Responsible AI and Risk Management

Study period

Aug 2014 - Aug 2018